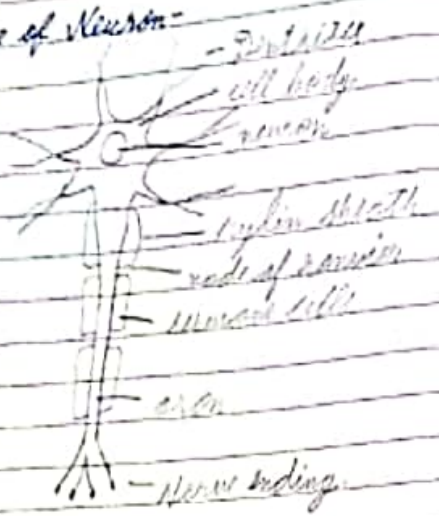
Control and coordination

Control -
When a body can manage all it's activities without disturbing the other factors such a condition is known as the body is ~~set~~ said to be able to control itself.

Coordination -
* When a different organ and organ system of the body coordinate and work in absolute integration with each other such a condition is known as coordination.
* Control and coordination are always in the coordinati to the given surrounding.

Animals-nervous system -
* In animals control and coordination is maintained by nervous and muscular tissues.
* Animal body can detect and respond to any emergency situation of the surrounding.
* The functional unit of nervous system is neuron.
* Neuron is capable to sense or detect surrounding environmental situation which is outside the body and can send message to the body.
* The corresponding body organ will recieve, read and react accordingly.

Structure of Neuron-



* The neuron is the functional unit of nervous system.
* Nerve cells receive information from surrounding, reads the information, process the information, and leads to a desired resultant.

* Information travels within the neuron in the form of electricity.

* Information which is not inside a neuron is present in the form of chemical nature.

* Information travels in a body in the form of electro-chemical Nature.

* Hence, it is said that Information travels in the form of electrical impulses.

* The space between two neuron is known as synapse or synaptic cliff or synaptic gap.

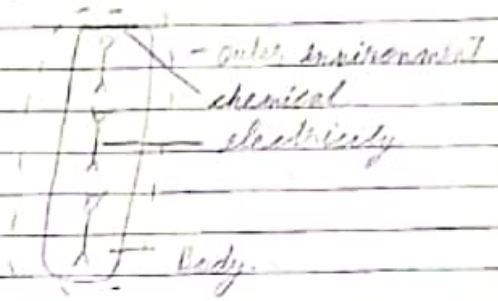
* Information travels within a Neuron unidirectionally from Dentrites to Nerve endings.

* Dentrites of the neuron receives information.

* Nerve endings of the neuron transmits the information.

* Dentrites receives the information in the form of chemical Nature.

* Nerve endings deliver the information in the form of Electrical impulses.



* Sensory organs parts of our body receives information from the surrounding.

Reflex action-

* The response given by the body in effect to the action done is known as Reflex arc.

* In reflex action brain does not depth takes any information nor processes it only Spinal chord ~~it involved~~ receives the information, reads the information processes and the information and takes required action.

The decision take by the spinal chord is stored in the brain.

Reflex Arc-

Receptor → sensory neuron → Spinal cord
↓
Effector muscle ← Motor Neuron ← Relay neuron

* receptor -
Sensory organs, which receives information from external environment.

* Sensory Muscle-
Neuron carrying information from receptor to spinal cord.

* Spinal chord-
- Receives information from sensory neuron.
- Processes the information.
- Transmits the processed information to relay neuron.
- Send the information and processed message to brain.

* Relay Neuron-
Receives message from spinal cord and transfers to neuron.

* Motor Neuron-
Carries information to the effector

* Effector-
Performs / Execute the action according to the processed information.

- Reflex action does not show any thinking mechanism.
- Spinal cord does not have thinking capabilities.
- Junction of neuron is called GANGLION.
- Cluster of gangliion is called GANGLIA.
- Hence, Brain can be considered as a big gangla.

HUMAN BRAIN-

* Brain is the junction of Neuron
* Brain participates in thinking.
* Thinking is a complex processes.
* Brain is the main co-ordinating centre of our body.
* Brain receives all the information and intergrates it.
* Brain controls all the voluntary actions.
* Brain sends processed information to muscles.
* Communication between CNS and other body parts is maintained by PNS

CNS ≡ Central Nervous System ≡ Brain + Spinal cord.

PNS ≡ Peripheral Nervous System ≡ Nerves arising from brain + Nerves arising from spinal cord.

* Brain is responsible for integration of input and output.
* Brain is divided into

Fore Brain | Mid Brain | Hind Brain

Fore Brain:
* Performs thinking
* recieves information from sensory organs.
* all the sites of sensory organ are different.
* It also stores pre-information.
* Decision is ~~not~~ taken by fore brain.
* It passes information to motor muscles.
* It controls voluntary actions.
* Feeling of hunger and fullness is controlled by fore brain.

Types of Action

Reflex Action	Involuntary action	voluntary action.
- They require an ~~of~~ external stimulus to initiate	- They are not under our control and do not require any stimuli	- actions which are under our control

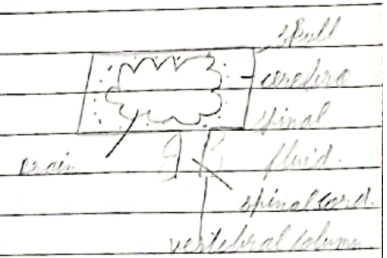
* Involuntary actions are controlled by mid-brain and hind brain.
* Involuntary actions like vomiting, salivation are controlled by hind brain.
* voluntary actions, where body balance and maintaining posture is controlled by hind brain.

Hind brain = cerebellum.

Tissues which protects CNS:
- Brain is a delicate organ
- Brain is protected by a bony box called skull / ~~sa~~ cranium
- Between skull and brain, there is a fluid to absorb shocks called cerebro-spinal fluid.
- Spinal chord is protected by vertebral column / backbone.

How does nervous tissue cause action? -

- Nervous system only supports in transfering information.
- The main action of a work is performed by muscle tissues.
- Muscles have contractile (~~protein~~) tissues.
- These tissues helps in contraction.
- Resulting in decrease in size of the muscle.
- Hence, the work is done.



Co-ordination in Plants -

* All plants are sensitive.
* Plants respond to stimuli.
* Plants respond towards stimuli

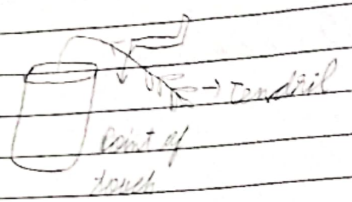
→ Imidiate response to stimuli

* Plants show different types of responses towards

various stimuli.

* Plants lack ~~to~~ nervous system and muscle system like animals.
* Plants responds towards various stimuli and shown trapism movement which may or may not be growth related.
* Plants ~~do not~~ communicate internally by electro-chemical processes.
* Plants do not have any special tissue for conduction of information.
* Plants cell changes its shape according to the changing amount of water in them. resulting in swelling and shrinking.

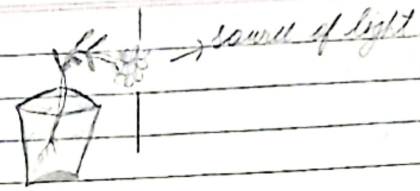
Movement due to growth-



* When we find movement in plants due to stimulus of touch in such cases the movement is growth related and the plant ~~to~~ body will move towards the direction of stimulus of touch such type of movement is known as - -

Thigma trapism
↓ touch ↓ movement.

* Phototrapism - movement seen in plants due to presence or absence of light.
↑ light ↑ to movement.



* Plants when ~~plants~~ placed in a dark room are seen to bend towards the source of light.
* The shoot system of the plant bends towards the light and the root system moves away from the light.

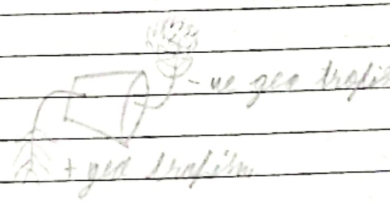
* chemotrapism - Movement shown by different parts of the plants towards or away from a chemical is known as chemotrapism
↓ chemical movement ↓

* Pollen grains after pollination produces pollen tube.
* These pollen tubes are always directed to the ovary or ovule of the plant.

Geotrapism.-

Geo - gravity , trapism - movement

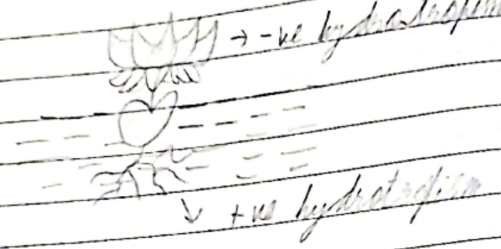
* movement of plants shown under the presence of gravity.



Hydrotropism -

Hydro / tropism
water / movement

* The movement in plants shown under the influence of water.



classical example of phototropism -

- During the course of time sunflower always face towards the SUN.

* Animals have control and directional growth. Eg. - our fingers don't grow haphazardly.

Hormones in Plants -

* Apart ~~fr~~ from tropic movement hormones also control plant movement.

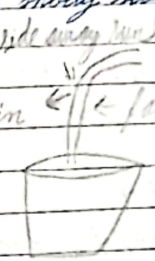
Hormones -

* Hormones are chemicals which are produced in traces with non-nutritive nature (very small amount).

* They help in growth, coordination and development of organism.

- different Hormones -

* Auxin - * It is a growth hormone.
  * Responsible for increase in size of the plants.
  * They are found on the shady side in case of phototropism.
  * helps in growth of leaves, stem, roots, branches.



* Gibberellin - * It is also a growth hormone.
  * Responsible for increase in growth of stem.

* Cytokinins - * It helps in cell division ~~and promotes~~
  * Promotes cell division in fruit and seed formation.

* Abscisic Acid - * It is an inhibiter. It checks for the growth of plants.
  * It causes wilting of leaves.

* Ethylene - * It promotes in ripening of fruits.

~~Hormones of in Animals~~

Hormones in Animals -

* Pituitary Gland →
Hormone produced - Growth ~~A~~ hormone.
location - Brain
function - Regulates growth in body

* Pineal Gland →
Hormone produced -
location - Hind Brain
function -

* Thyroid →
Hormone produced - Thyroxin.
location - throat
function - controls metabolism.

* Hypothalamus →
Hormone produced -
location - hid brain
function -

* Thymus →
Hormone produced - Thymosin
location - behind ~~section~~ behind sternum
function - ~~to~~ controls immunity.

* Pancrease →
Hormone produced - Insulin
~~to~~ location - Near Stomach
function - controls blood sugars.

* Adrenal gland -
Hormone produced - Adrenaline / flight or fight hormone
location - above ~~to~~ kidney.
function - Helps to take quick acts at emergency ~~&~~ situa

* Ovary -
Hormone produced - Estrogen + Progestron.
location - Ovary
function - controls primary and secondary sexual characte

* Testis -
Hormone produced - Testerone
location - Testis.
function - controls primary & secondary sexual characte